Probing Ultrafast Photocurrents

Investigating hot-carrier dynamics in Silicon single crystals using photocurrent spectroscopy

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Motivation & Background

- **Hot carrier-dynamics:** Very little known about hot state conductivity.
- **Photocurrent spectroscopy:** Measuring carrier-dynamics with pico-second resolution.
- **Shockley-Quaser:** Thermodynamical limiting efficiency for solar panels.

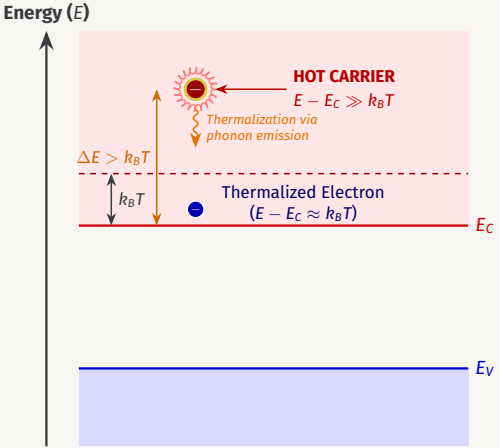
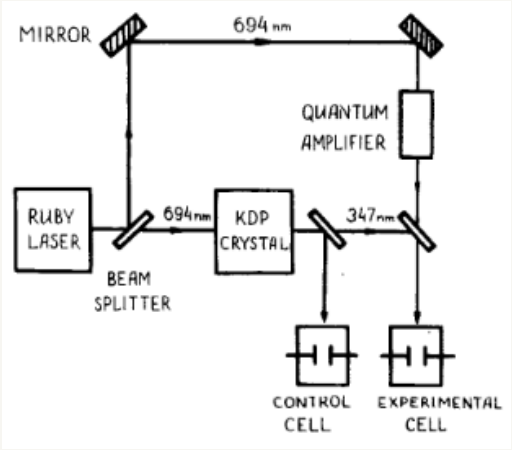


Photo- current/ conductivity Spectroscopy

- Optical pump electrical probe
- Electrical detection
- Advantages over TA methods



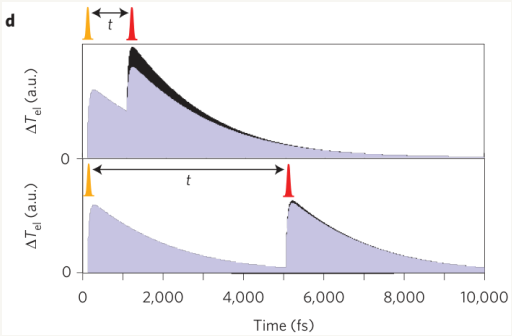
L.V Lukin, *et al*, Chemical Physics Letters, Volume 81, Issue 3, 1981.

Experimental Methods

Conductivity changes

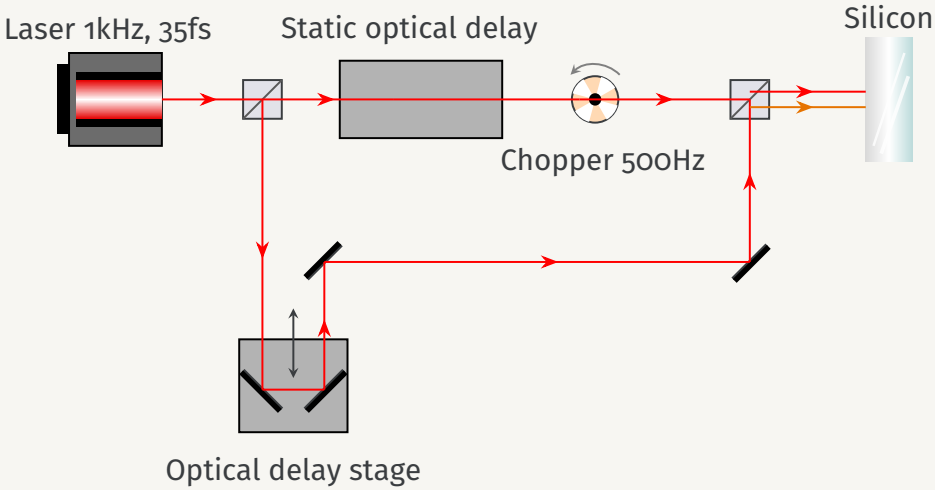
The material responds to illumination by changing its conductivity:

$$\Delta\sigma = \Delta\mu\Delta N \tag{1}$$



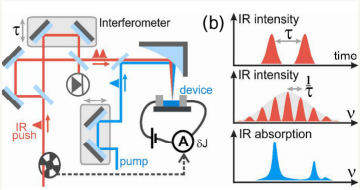
Tielrooij, K. J. *et al.* Nature Nanotech 10, 437–443 (2015).

Experimental Setup

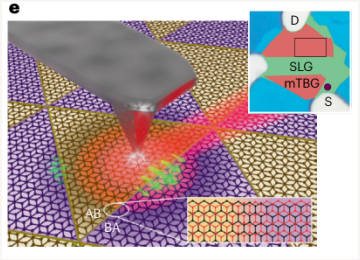


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- Correlation spectroscopy
- Vibrationally controlled spectroscopy
- Near field



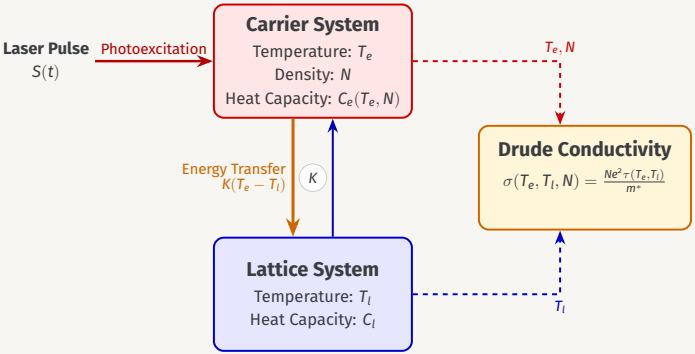
Bakulin, A. A. *et al.* Nat Commun, 7880 (2015).



Qiong Ma, *et al.* Nature Reviews Physics 5, 170-184 (2023).

Carrier Two-Temperature Model

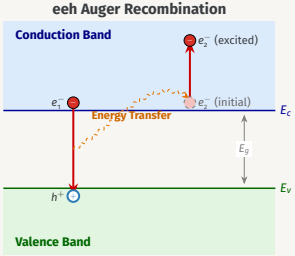
- Carrier population
- Carrier temperature
- Lattice temperature
- Conductivity

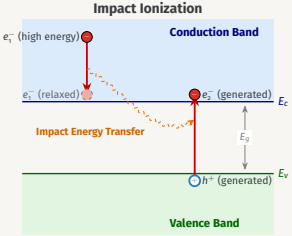


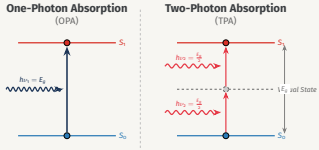
Carrier Two-Temperature Model

Carrier Population Model

$$\frac{\partial n}{\partial t} = -\gamma n^3 + \eta n + \frac{\alpha}{\hbar\omega_0} I + \frac{\beta}{\hbar\omega_0} I^2 \tag{2}$$





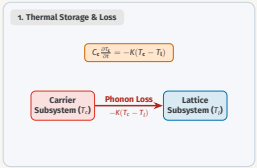


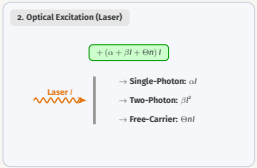
Carrier Two-Temperature Model

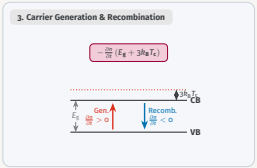
Carrier Temperature Model

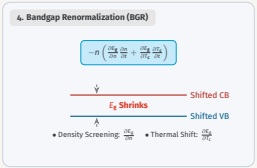
Carrier temperature model

$$C_c \frac{\partial T_c}{\partial t} = -K\left(T_C - T_l\right) + \left(\alpha + \beta I + \Theta n\right) I - \frac{\partial n}{\partial t}\left(E_g + 3k_B T_c\right) - n\left(\frac{\partial E_g}{\partial n}\frac{\partial n}{\partial t} + \frac{\partial E_g}{\partial T_c}\frac{\partial T_c}{\partial t}\right) \quad (3)$$







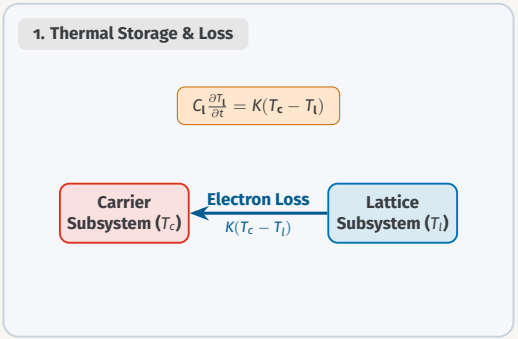


Carrier Two-Temperature Model

Lattice Temperature Model

Lattice temperature model

$$C_l \frac{\partial T_l}{\partial t} = K\,(T_c - T_l) \tag{4}$$



Carrier Two-Temperature Model

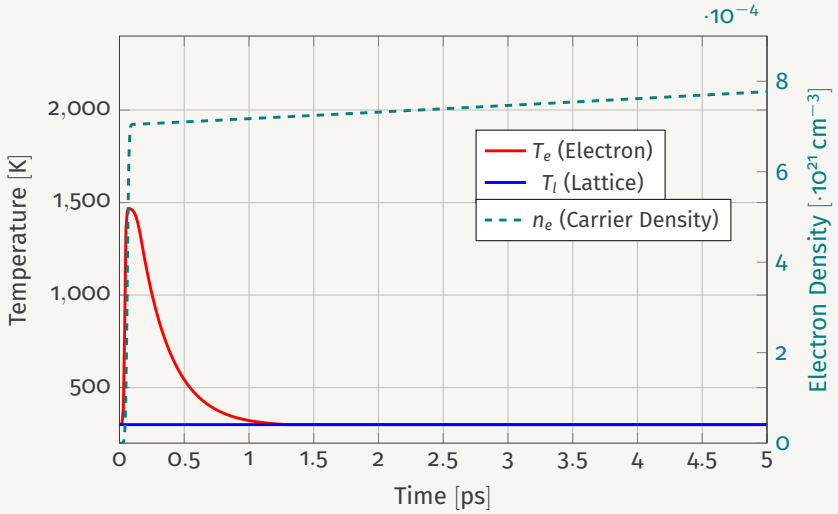
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Drude Model

Drude conductivity

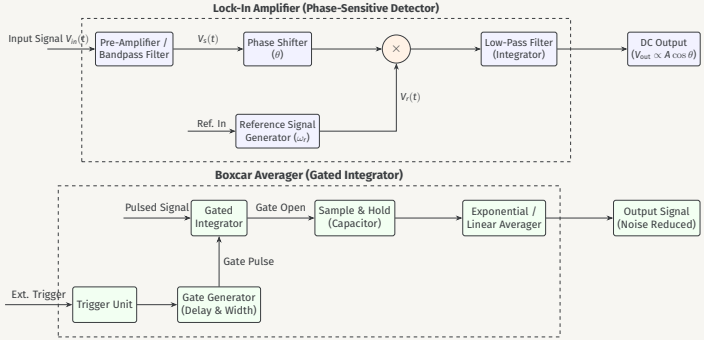
$$\sigma = \frac{\sigma_0}{\nu\left(T_l\right)} \tag{5}$$

Carrier Two-Temperature Model

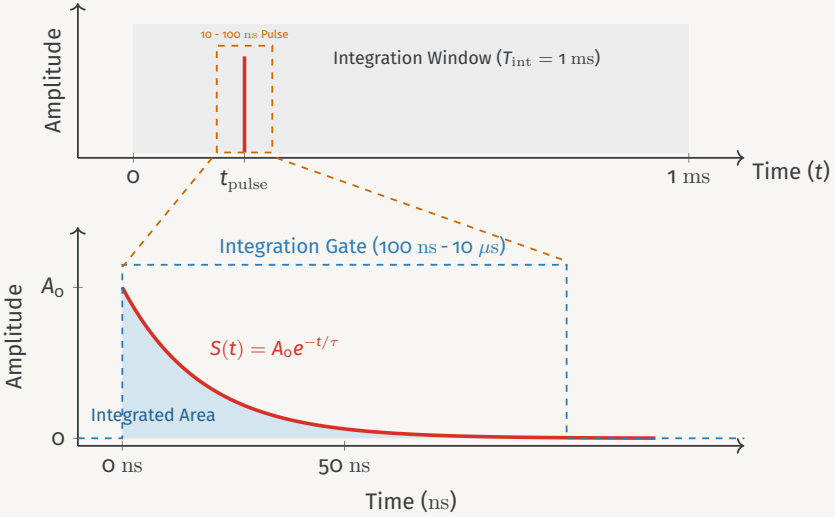


Electrical Detection

- 500 Hz modulation
- 1 kHz repetition rate
- Lock-in amplifier SR-830
- Box car averager



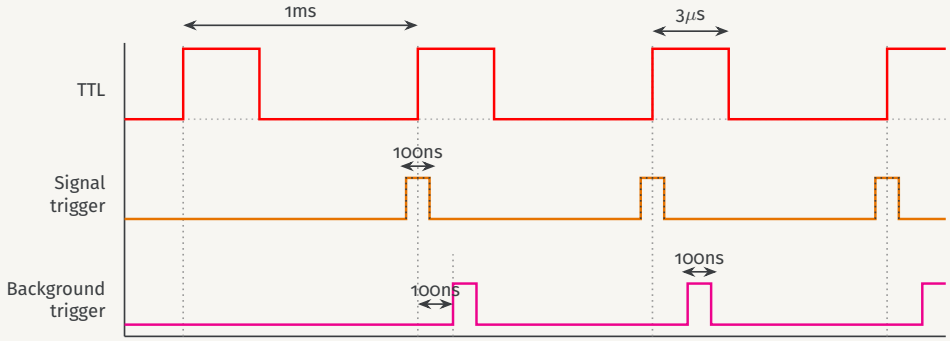
Electrical Detection



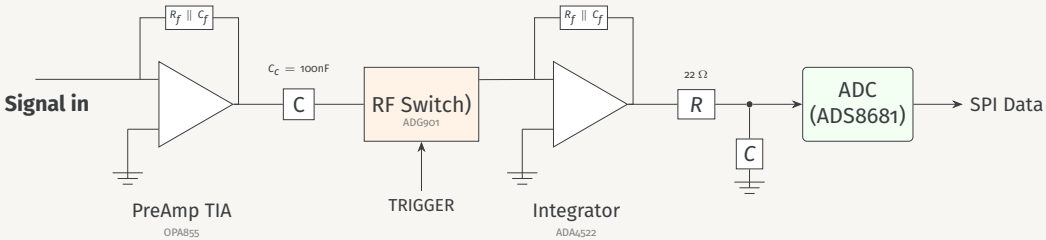
Electrical Detection

Box car

$$A[k] = \frac{1}{N} \sum_{i=0}^{N-1} \left(Signal\,[i] - Background\,[i] \right). \tag{6}$$

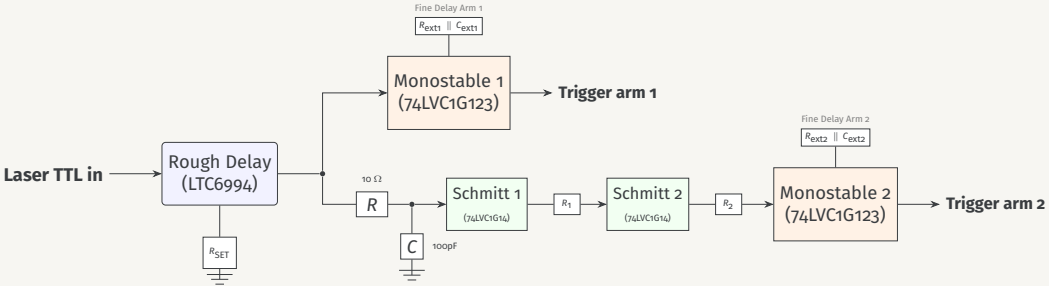


Electrical Detection



Component	Bandwidth	Noise
Pre-amp	8 GHz	0.5 pA/$\sqrt{\text{Hz}}$
Gated integrator	4 GHz	2.5 nV/$\sqrt{\text{Hz}}$

Electrical Detection



Trigger stage	Imparted delay	Phase jitter [ppm/Hz]	Pulse duration
Rough delay	1 ms	7 ppm	1 µs